

Chapter 5 : Markov Chain Monte Carlo

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Introduction

We have seen several methods to simulate independent and identically distributed random variables $\{X_1, X_2, \dots\}$ according to a given distribution f . Assuming this is feasible, the chapter on Monte Carlo methods presented a classical application to the estimation of

$$I = E[g(X)] = \int g(x)f(x)dx,$$

first through the standard Monte Carlo estimator

$$\hat{I}_n = \frac{1}{n} \sum_{i=1}^n g(X_i).$$

Introduction

The convergence of $\hat{I}_n$ to I remains valid even if the variables X_n are not independent., but form a Markov chain $X_1, X_2, \dots$ with stationary distribution f .

Remark: For practical purposes, we consider a finite state space, but all results can be extended to a more general setting.

This is known as **Markov Chain Monte Carlo (MCMC)**, of which the Metropolis–Hastings algorithm is the most well-known example.

Markov Chain Monte Carlo Method

This is a method for generating a sequence $\{X_1, X_2, \dots\}$ of random variables that can be interpreted as constituting a trajectory of a Markov chain over M states, possessing the stationary distribution $\pi = (\pi_1, \dots, \pi_M)$ with $\pi_i \geq 0$ and $\sum_i \pi_i = 1$ for all i .

This method proves particularly rich in applications ranging from generating states of physical systems in thermodynamic equilibrium to obtaining random points in complicated regions.

Markov Chain Monte Carlo Method

Based on an analogy with the annealing process (common in metallurgy, it consists of heating and then very slowly cooling an alloy to enable it to reach a state that minimizes its internal energy), the Cerny-Kirkpatrick method is known as simulated annealing.

It is also widely used in the learning phase of certain artificial neural networks, as discussed in particular in Aarts and Korst's book on Boltzmann machines.

Markov Chain Monte Carlo Method

It should now be clear what the general strategy of the MCMC technique is: given a desired distribution π on the sample space S , find a (preferably aperiodic) irreducible transition matrix P that has π as stationary distribution, and apply the ergodic theorem to estimate expectation values. Such a transition matrix is far from unique: if the sample space has M states, there are many transition matrices that share the same stationary distribution. So we need some guiding principle in constructing transition matrices. A particularly convenient principle is that of detailed balance.

Detailed balance

Note that the stationarity condition $\pi P = \pi$ is equivalent to requiring the following identity for all $j \in S$

$$\sum_{i=1}^M \pi_i \pi_{ij} = \pi_j = \pi_j \sum_{i=1}^M \pi_{ji} = \sum_{i=1}^M \pi_j \pi_{ji}.$$

The left-hand side gives the total flow of probability from any state i into j , while the right-hand side gives the flow out of j to any state i . In equilibrium, i.e. in stationary distribution, these two flows must be equal.

Detailed balance

The simple way to ensure the balance condition is to require detailed balance,

Definition

Let $\pi = (\pi_1, \dots, \pi_M)$ a probability distribution and (X_n) Markov chain with transition matrix P . We will say that π is **reversible** for the chain if and only if it verifies the **detailed equilibrium** equations:

$$\pi_i p_{ij} = \pi_j p_{ji}, \forall 1 \leq i, j \leq M. \quad (1)$$

Meaning that not just the total flow in and out of state j is balanced, but that the flow between any two states i and j is the same in both directions.

Reversibility

Example

Suppose we have an undirected graph with vertices numbered from 1 to M and for each vertex $i \in \{1, \dots, M\}$, let d_i be the number of edges connected to it. Now consider the following Markov chain: starting from vertex i , we move to one of the d_i neighboring vertices with equal probability. It is easy to verify that the measure

$$\pi = \left[\frac{d_1}{\sum_{j=1}^M d_j}, \dots, \frac{d_M}{\sum_{j=1}^M d_j} \right]$$

is reversible for this chain. This random walk is one of the ingredients of Google's famous PageRank algorithm, with the vertices corresponding to web pages and the edges to the links between them.

Reversibility

Lemma (reversibility $\Rightarrow$ stationarity)

Let (X_n) be a Markov chain with transition matrix P . If the measure π is reversible for the chain, then it is invariant, i.e., $\pi P = \pi$.

Remark

Among Markov chains, reversible chains are those that will be of interest to us going forward.

Metropolis Algorithm

We first present the algorithm proposed by Metropolis and his co-authors in 1953, before moving on to its generalization by Hastings in 1970. We want to simulate a Markov chain with a stationary distribution π , possibly known only to within one normalization constant :

$$\forall j \in \{1, \dots, M\} \quad \pi_j = C \pi_j^u = \frac{\pi_j^u}{\sum_{i=1}^M \pi_i^u}, \quad (2)$$

where π^u stands “ π unnormalized”.

This problem is anything but artificial: it is, for example, recurrent in Bayesian statistics and statistical physics. For simplicity, we'll assume that $\pi_j > 0$ for all j .

Metropolis Algorithm

To construct this chain, we need to specify an irreducible and aperiodic transition matrix P with π as its stationary law. Like Acceptance-Rejection and Importance sampling methods. The idea is to use the proposal auxiliary (or instrumental) mechanism. Let an irreducible transition matrix $Q = [q_{ij}], 1 \leq i, j \leq M$ such that:

- Symmetry : for all pair (i, j) , we have $q_{ij} = q_{ji}$;
- Starting from any state i , we can easily simulate according to the law $q_{i.} = [q_{i1}, \dots, q_{iM}]$.

Metropolis Algorithm

Algorithm

- 1 Starting from $X_n = i$, simulate $Y = j$ according to the law $q_{i\cdot} = [q_{i1}, \dots, q_{iM}]$;
- 2 Calculate the acceptance ratio $r_{ij} = \frac{\pi_j}{\pi_i}$;
- 3 Generate a uniform distribution $U \sim \mathcal{U}[0, 1]$, and set

$$X_{n+1} = \begin{cases} Y = j & \text{if } U \leq r_{ij}, \\ X_n = i & \text{otherwise.} \end{cases}$$

Metropolis Algorithm

Remark: When constructing the sequence (X_n) we must conserve everything, including the times when we remain in place (i.e. $X_{n+1} = X_n$).

Remark

The advantage of this algorithm is that it only uses the ratios $\frac{\pi_j}{\pi_i} = \frac{\pi_j^u}{\pi_i^u}$. It therefore does not require knowledge of the normalization constant C in formula 2.

Proposition (Metropolis Reversibility) If the transition matrix Q is irreducible, then the chain (X_n) produced by Metropolis algorithm is irreducible and π is reversible for this chain. In particular, π is its unique balance distribution.

Metropolis Algorithm

Example

Consider the state space: $S = \{0, 1, 2, 3, 4\}$ and the target distribution:

i	0	1	2	3	4
π_i	0.05	0.10	0.50	0.20	0.15

Assume that we cannot sample directly from this distribution.

- 1 Give the proposal matrix Q using the following mechanism:
At each step, move one position to the left or to the right.
- 2 For each possible transition, compute the acceptance probabilities $r_{ij} = \min\left(1, \frac{\pi_j}{\pi_i}\right)$.
- 3 Compute the transition matrix $P [p_{ij} = q_{ij}r_{ij}]$.
- 4 Verify the stationarity of π .

Metropolis-Hastings Generalization

In the previous algorithm, the dynamics defined by the transition matrix Q were assumed to act symmetrically on the state space, i.e.

$$q_{ij} = q_{ji} \quad \forall (i, j) \in S \times S.$$

In applications, this condition is not necessary. To get away from this, Hastings proposed an approach that makes it possible to start from a matrix Q of an irreducible Markov chain. But all that needs to be taken into account is the possible asymmetry of Q in the ratio r_{ij} . This is how Hastings generalized the Metropolis algorithm.

Metropolis-Hastings Generalisation

Metropolis-Hastings Algorithm

For this generalization is simply to assume that:

- starting from any state i , we can easily simulate according to the law $q_i = [q_{i1}, \dots, q_{iM}]$.
- starting from any state (i, j) : $q_{ij} > 0$ implies $q_{ji} > 0$;
- starting from any state (i, j) such that $q_{ij} > 0$, we know how to calculate $\frac{\pi_j q_{ji}}{\pi_i q_{ij}}$.

Generalization

Metropolis-Hastings Algorithm

Algorithm

- 1 Starting from $X_n = i$, simulate $Y = j$ according to the law $q_i = [q_{i1}, \dots, q_{iM}]$;
- 2 Calculate the Metropolis-Hastings ratio

$$r_{ij} = \frac{\pi_j q_{ji}}{\pi_i q_{ij}},$$

- 3 Generate a uniform distribution $U \sim \mathcal{U}[0, 1]$, set

$$X_{n+1} = \begin{cases} Y = j & \text{if } U \leq r_{ij}, \\ X_n = i & \text{otherwise} \end{cases}$$

Continuous State Space

Proposition (Metropolis-Hastings Reversibility) *If the transition matrix Q is irreducible and aperiodic, the chain (X_n) produced by the Metropolis algorithm is irreducible, aperiodic and π is reversible for this chain.*

Furthermore, everything that has been said in the case of a finite state space is generalized to an infinite state space, whether countable or continuous. In the case of a **continuous state space**, let's say $\mathbb{R}^d$, the aim is to simulate according to the density

$$f(x) = Cf_u(x) = \frac{f^u(x)}{\int f^u(y) dy}, \quad (3)$$

where, as in formula 2, f^u means “ f unnormalized”.

Continuous State Space

The transition matrix $Q = [q_{ij}]_{1 \leq i, j \leq M}$ replaced by a **transition kernel** $q(x, y)$, i.e: $q(x, y) \geq 0$ for all pair (x, y) and

$$\forall x \in \mathbb{R}^d, \quad \int q(x, y) dy = 1.$$

Like its discrete counterpart, the kernel q can be seen as a way of moving through the state space $\mathbb{R}^d$. With the variable x fixed, the quantity $q(x, y)$ can be interpreted as the probability density of going to y knowing that we start from point x .

Metropolis-Hastings Generalization

Metropolis-Hastings Algorithm, continuous case

The assumptions required for the Metropolis-Hastings algorithm in the continuous case are then:

- for any points x , we know how to simulate according to the density $q(x, \cdot)$;
- for any pair (x, y) , $q(x, y) > 0$ implies $q(y, x) > 0$;
- for any pair (x, y) such that $q(x, y) > 0$, we can calculate $\frac{f(y)q(y, x)}{f(x)q(x, y)}$.

Continuous State Space

Metropolis-Hastings Algorithm, continuous case

Algorithm

- 1 Starting from $X_n = x$, simulate $Y = y$ according to the law $q(x, \cdot)$;
- 2 Calculate the Metropolis-Hastings ratio

$$r(x, y) = \frac{f(y)q(y, x)}{f(x)q(x, y)}$$

- 3 Generate the uniform distribution $U \sim \mathcal{U}[0, 1]$,
set $X_{n+1} = Y = y$ if $U \leq r(x, y)$, otherwise $X_{n+1} = X_n = x$.

Under reasonable assumptions about the transition kernel q , the chain (X_n) has the unique stationary law f .

Continuous State Space

Example

We want to simulate a real-valued random variable $X \in \mathbb{R}$ with target density

$$f(x) = e^{-x^4 + 2x^2}$$

- 1 Use a random walk proposal to define the proposal matrix Q .
- 2 Show that the acceptance probability reduces to $r(x, y) = \min\left(1, \frac{f(y)}{f(x)}\right)$, and give the explicit formula.
- 3 Write the Metropolis-Hastings algorithm for this problem.
- 4 Explain what happens when:
 - σ is too small,
 - σ is too large.

Gibbs Sampler

Boltzmann-Gibbs Distribution

Let the state space S be finite (very large), and suppose we want to minimize a function $V : S \rightarrow \mathbb{R}$ such that $V(x)$ represents the **energy** associated with the state $x \in S$ of the system. Then, under fairly general conditions, the stationary distribution corresponding to the thermodynamic equilibrium of the system's states is a **Boltzmann-Gibbs** law:

$$f_T(x) = \frac{1}{Z_T} \exp\left(-\frac{V(x)}{kT}\right), x \in S \quad (4)$$

also known as the Gibbs measure associated with the potential of V and the temperature T , where k represents the Boltzmann constant and Z_T is the normalization constant, called the partition function in statistical physics, it is defined by:

$$Z_T = \sum_{x \in S} \exp\left(-\frac{V(x)}{kT}\right).$$

Gibbs Sampler

Boltzmann-Gibbs Distribution

We assume that Boltzmann's constant has been incorporated into the temperature, or what amounts to the same thing, we'll set $k = 1$ in the relation (4) as well as in the distribution function. The equation then becomes

$$f_T(x) = \frac{1}{Z_T} \exp\left(-\frac{1}{T} V(x)\right), x \in S. \quad (5)$$

with

$$Z_T = \sum_{x \in S} \exp\left(-\frac{1}{T} V(x)\right).$$

Depending on its value, temperature T tends to accentuate or smooth out the extrema of V .

Boltzmann-Gibbs Distribution

Example

Let $S = \{1, 2, 3\}$, with $V(1) = 0, V(2) = 1, V(3) = -1$.

One can easily verify the following two extreme behaviors:

$$\lim_{T \rightarrow 0^+} f_T(x) = [0, 0, 1] \text{ and } \lim_{T \rightarrow +\infty} f_T(x) = \left[\frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right].$$

In particular, as the temperature tends to zero, the Gibbs measure concentrates on the minimizer of the potential V .

This result is generalized for infinite or uncountable state space S . Let us study the two limiting cases obtained when T tends to ∞ and 0 , respectively.

Gibbs Sampler

Simulated Annealing

When $T \rightarrow \infty$, we obtain

$$\lim_{T \rightarrow \infty} f_T(x) = \lim_{T \rightarrow \infty} \frac{\exp(-V(x)/T)}{\sum_{x \in S} \exp(-V(x)/T)} = \frac{1}{|S|}, \forall x \in S$$

which is nothing but the uniform distribution.

When $T \rightarrow 0$, let

$$S_0 = \{x \in S : V(x) \leq V(y), \forall y \in S\},$$

then

$$\lim_{T \rightarrow 0} f_T(x) = \lim_{T \rightarrow 0} \frac{\exp(-V(x)/T)}{\sum_{y \in S} \exp(-V(y)/T)} = \begin{cases} \frac{1}{|S_0|}, & x \in S_0 \\ 0, & x \in S \setminus S_0 \end{cases}.$$

Gibbs Sampler

Simulated Annealing

After introducing the reciprocal of the numerator term into the sum in the denominator and using

$$\lim_{T \rightarrow 0} \exp \left(-\frac{V(y) - V(x)}{T} \right) = \begin{cases} 1, x \in S_0, y \in S_0 \\ 0, x \in S_0, y \notin S_0 \end{cases} ,$$

we conclude that, as the temperature tends to zero, the distribution allocates all its probability mass to the states of minimal energy.

Gibbs Sampler

Simulated Annealing

- Since at low temperatures, the Gibbs measure focuses on global minima of V , one idea is to give ourselves a temperature sequence (T_n) decrising to 0 and for all n , simulate a Markov chain (X_n) that has f_{T_n} as its stationary law.
- For a given transition matrix $Q(x, y)$ and verifying the properties required by Metropolis-Hastings (irreducibility and aperiodicity), we then have the following algorithm, called **simulated annealing** (coming from metallurgy).

Gibbs Sampler

Metropolis Algorithm for Simulated Annealing

Algorithm

- 1 Starting from $X_n = x$, simulate $Y = y$ according to the law $Q(x, \cdot)$;
- 2 Calculate the Metropolis-Hastings ratio

$$\begin{aligned} r_n(x, y) &= \frac{f_{T_n}(y) Q(y, x)}{f_{T_n}(x) Q(x, y)} \\ &= \frac{Q(y, x)}{Q(x, y)} \times \exp \left(-\frac{1}{T_n} (V(y) - V(x)) \right), \end{aligned}$$

- 3 Generate the uniform distribution $U \sim \mathcal{U}[0, 1]$,
Set $X_{n+1} = Y = y$ if $U \leq r_n(x, y)$, otherwise
 $X_{n+1} = X_n = x$.

Gibbs Sampler

Simulated Annealing

Remark :

- 1 The chain is no longer homogeneous due to temperature changes T_n .
- 2 If the transitions are symmetrical, i.e. $Q(x, y) = Q(y, x)$, the Metropolis algorithm is applied with the ratio

$$r_n(x, y) = \exp \left(-\frac{1}{T_n} (V(y) - V(x)) \right),$$

which is greater than or equal to 1 whenever $V(y) \leq V(x)$.

- 3 To avoid getting trapped in a local minimum V , we systematically accept a transition to a lower value of V , but we don't systematically refuse a transition to a higher value.
- 4 We can also see that, for fixed x and y such that $V(y) > V(x)$, as the temperature is lower, we will be less inclined to accept a transition to a higher value.